- Isolation from host machine
- Easy scaling and orchestration with Kubernetes or Docker Compose
- Fast and reproducible deployments

17.6.2 Installing Docker

To install Docker on your local machine or server, follow the official Docker installation guide for your operating system. You can find the instructions here: Docker Installation Guide, https://docs.docker.com/engine/install/.

17.7 Exercise 50: Deploying Laravel 12 to Production on Ubuntu Server with PHP 8.4

17.7.1 Description

In this lab, we will deploy a Laravel 12 application on a production-ready environment using a virtual machine running **Ubuntu Server** and **PHP 8.4.x**. You'll configure Nginx, PHP-FPM, MySQL, and deploy Laravel from a GitHub repository.

17.7.2 bjectives

By the end of this lab, you will be able to:

- Set up a production server with PHP 8.4 for Laravel 12
- Install and configure Nginx, PHP-FPM, Composer, and MySQL
- Deploy and configure Laravel application securely on Ubuntu

17.7.3 Prerequisites

- Laravel 12 project (GitHub or ZIP)
- A VM running **Ubuntu Server 22.04 or later**
- SSH access to the server
- (Optional) Domain name pointing to the server

17.7.4 Steps

Here's a step-by-step guide to deploying Laravel 12 on Ubuntu Server with PHP 8.4.